

# Bentley Oakes

Assistant Professor

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## Key Academic Skills

**Researcher:** Lead- and co-author of dozens of publications with hundreds of citations on Google Scholar.

**Collaborator:** Involved in multiple international academic and industrial collaborations.

**Teacher/Mentor:** Extensive experience teaching courses and mentoring graduate students.

**Community-Builder:** Organizing and program committee member for local and global events.

## Language Skills

**English:** Native proficiency

**French:** Conversational (improving)

## Education

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| <b>PhD</b>       | <b>McGill University</b> , Computer Science                                                                                                                                                                                    | Montréal, Canada |
|                  | <ul style="list-style-type: none"> <li>Thesis: <i>A Symbolic Execution-Based Approach To Model Transformation Verification using Structural Contracts</i></li> <li>Supervisors: Hans Vangheluwe and Clark Verbrugge</li> </ul> | Jan 2019         |
| <b>MSc</b>       | <b>McGill University</b> , Computer Science                                                                                                                                                                                    | Montréal, Canada |
|                  | <ul style="list-style-type: none"> <li>Thesis: <i>Practical and Theoretical Issues of Evolving Behaviour Trees for a Turn-Based Game</i></li> <li>Supervisor: Clark Verbrugge</li> </ul>                                       | Jan 2013         |
| <b>BSc</b>       | <b>University of Manitoba</b> , Computer Science                                                                                                                                                                               | Winnipeg, Canada |
| <b>(Honours,</b> | <ul style="list-style-type: none"> <li>Co-op internship at Electronic Arts Inc., Montréal — Fall 2009, Summer 2010: Prototyping artificial intelligence in commercial video games.</li> </ul>                                  | Jan 2011         |
| <b>Co-op)</b>    | <ul style="list-style-type: none"> <li>Co-op internship at Blackberry Limited (RIM), Waterloo — Winter 2009: Implementing cryptographic communication protocols.</li> </ul>                                                    |                  |

## Research Experience

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| <b>Polytechnique Montréal</b> , Assistant Professor      | <ul style="list-style-type: none"> <li>Department of Computer and Software Engineering (GIGL)</li> <li>Research topics: Accelerating the construction and deployment of digital twins through ontologies for capturing domain knowledge, and large language models and agentic AI for automating their creation. Integrating formal methods to verify digital twin correctness, and assisting users with the configuration and observability of their machine learning and verification/validation services.</li> </ul> | Montréal, Canada<br>Sept 2023 – present  |
| <b>Université de Montréal</b> , Post-Doctoral Researcher | <ul style="list-style-type: none"> <li>Lab: GEODES Software Engineering Research Group</li> <li>Supervisors: Houari Sahraoui and Michalis Famelis</li> <li>Research topic: Assisting non-machine learning experts in constructing ML solutions by exploring tailoring of computational workflows.</li> <li>Research topic: Extracting and structuring developer rationale from open-source commit messages using NLP and Large Language Models, in collaboration with PhD student Mouna Dhaouadi.</li> </ul>            | Montréal, Canada<br>Sept 2021 – Aug 2023 |

**University of Antwerp**, Post-Doctoral Researcher

- Labs: Antwerp Systems and Software Modelling, and Constrained Systems Lab
- Supervisors: Hans Vangheluwe and Joachim Denil
- Research topics: Verification and validation of cyber-physical systems, model-driven engineering, multi-paradigm modelling, co-simulation, and digital twins.

Antwerp, Belgium  
Sept 2018 – July 2021

**Université de Montréal**, Visiting Researcher

- Host: Eugene Syriani, GEODES Software Engineering Research Group
- Research topic: Developing an interface between the AToMPM modelling tool and the ModelVerse modelling repository.

Montréal, Canada  
May 2018

**fortiss GmbH**, Visiting Researcher

- Host: Levi Lúcio
- Research topic: Formalizing representations of model transformation languages.

Munich, Germany  
July 2016 – Aug 2016

**General Motors Technical Center**, Visiting Researcher

- Host: Ramesh Sethu
- Research topics: Applying model transformations for code/model modernization at an industrial scale, and industrial intellectual property concerns.

Warren, USA  
Oct 2014 – Dec 2014

## Industrial Collaborations

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**Framework for Systematic Design of Digital Twins (DTDesign)**

2019–2021

- Collaborators: Atlas Copco, Flanders Make
- Topics: Digital twin characteristics, integrating knowledge graphs and digital twins.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

**Automated & Simulation-based Functional Safety Engineering Methodology (aSET)**

2018–2020

- Collaborators: Dana Belgium NV, Siemens Industry Software (Leuven), Flanders Make
- Topics: Verification of safety-critical cyber-physical systems (formal methods, reinforcement learning-based fault injection for hazard discovery, simulation/visualization), DevOps for functional safety.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

**Innovation in the Development of Electrical Systems For Aeronautics (INES)**

2019–2020

- Collaborators: Boeing Research & Technology Europe (Madrid), Siemens Industry Software (Leuven), Flanders Make
- Topics: Co-simulation configuration, fault injection in co-simulation.
- Funding: Flanders Innovation and Entrepreneurship Agency (VLAIO)

**Network for the Engineering of Complex Software-Intensive Systems for Automotive Systems (NECSIS)**

2013–2016

- Collaborators: General Motors of Canada Ltd., Queen's University, University of Antwerp
- Topic: Verification of model transformations.
- Funding: \$14M from Automotive Partnership Canada and Natural Sciences and Engineering Research Council of Canada (NSERC)

## Teaching Experience

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**Polytechnique Montréal**, Course Instructor

- LOG6310E
- Engineering of Digital Twins — Winter 2025
- LOG8371E
- Software Quality Engineering — Fall 2024
- INF6900AE/INF7900A
- Scientific and Technical Communication — Winter 2024, Fall 2024

Montréal, Canada  
Jan 2024 – present

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| <b>Polytechnique Montréal</b> , Guest Lecturer <ul style="list-style-type: none"> <li>• LOG8704</li> <li>• XR Software Development — Nov. 2024: <i>Digital Twins: Introductions and Intersections</i></li> <li>• LOG6953DE</li> <li>• Model-Driven Software Engineering — Nov. 2022, Nov. 2023: Model-driven engineering, usage and verification of model transformations</li> </ul>                                                         | Montréal, Canada<br>Jan 2022 – Jan 2024 |
| <b>McGill University</b> , Course Lecturer/Coordinator <ul style="list-style-type: none"> <li>• COMP 202</li> <li>• Foundations of Programming — Winter 2015, 2017, 2018 (six terms, avg. 189 students)</li> <li>• Developed and presented lecture material on Java programming for students with no prior programming experience.</li> <li>• Created assessments and marking guides; coordinated with other instructors and TAs.</li> </ul> | Montréal, Canada<br>Jan 2015 – Jan 2018 |
| <b>University of Antwerp</b> , Teaching Assistant <ul style="list-style-type: none"> <li>• 2001WETMTR</li> <li>• Model-Driven Engineering — Fall 2020</li> <li>• 2001WETMSI</li> <li>• Modelling of Software-Intensive Systems — Fall 2019</li> </ul>                                                                                                                                                                                        | Antwerp, Belgium<br>Jan 2019 – Jan 2020 |
| <b>McGill University</b> , Teaching Assistant <ul style="list-style-type: none"> <li>• COMP 202</li> <li>• Foundations of Programming (×2)</li> <li>• COMP 250</li> <li>• Introduction to Computer Science (×2)</li> <li>• COMP 251</li> <li>• Data Structures and Algorithms (×3)</li> </ul>                                                                                                                                                | Montréal, Canada<br>Jan 2012 – Jan 2014 |

## Invited Talks and Presentations

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| <b>Guest lecture — Engineering Digital Twins course, McMaster University</b> <ul style="list-style-type: none"> <li>• Host: Prof. Istvan David</li> <li>• <i>What Does Your Digital Twin Do? A Framework and Tooling for Systematic DT Reporting</i></li> </ul> | Mar. 2026 |
| <b>Invited talk — SIG LLODIA (Lecture #3)</b> <ul style="list-style-type: none"> <li>• <i>Towards Ontology-based Digital Twin Service Construction and Reporting</i></li> </ul>                                                                                 | Jan. 2026 |
| <b>Guest lecture — RAISE course, UCLA</b> <ul style="list-style-type: none"> <li>• Host: Dr. Maged Elaasar</li> <li>• <i>What is a 'Digital Twin' and How Do I Build One?</i></li> </ul>                                                                        | Nov. 2025 |
| <b>Tutorial — Software Engineering for Machine Learning Applications (SEMLA)</b> <ul style="list-style-type: none"> <li>• <i>What is a 'Digital Twin' and How Do I Build One?</i></li> </ul>                                                                    | Jun. 2025 |
| <b>Invited talk — EDT.Community</b> <ul style="list-style-type: none"> <li>• <i>Systematic Digital Twin Reporting</i></li> <li>• <a href="#">Video recording</a></li> </ul>                                                                                     | May 2025  |
| <b>Invited talk — Next-Generation Cities Institute, Concordia University</b> <ul style="list-style-type: none"> <li>• <i>Accelerating Digital Twin Research and Engineering</i></li> </ul>                                                                      | May 2025  |
| <b>Conference paper presentation — ICSE 2025, Ottawa</b> <ul style="list-style-type: none"> <li>• <i>Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice</i> (journal-first)</li> </ul>                   | May 2025  |

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| <b>Guest lecture — Engineering Digital Twins course, McMaster University</b> <ul style="list-style-type: none"> <li>• Host: Prof. Istvan David</li> <li>• <i>Accelerating Digital Twin Research and Engineering</i></li> </ul>                        | Mar. 2025 |
| <b>Research presentation — KARI Delegation, Polytechnique Montréal</b> <ul style="list-style-type: none"> <li>• Advanced Air Mobility (AAM) challenges and advancements</li> </ul>                                                                    | Feb. 2025 |
| <b>Invited talk — Software Engineering at Montréal (SEMTL)</b> <ul style="list-style-type: none"> <li>• <i>Recent Work on Systematic Reporting and Construction for Digital Twins</i></li> </ul>                                                      | Jan 2025  |
| <b>Guest lecture — LOG8704 XR Software Development, Polytechnique Montréal</b> <ul style="list-style-type: none"> <li>• <i>Digital Twins: Introductions and Intersections</i></li> </ul>                                                              | Nov. 2024 |
| <b>Conference paper presentation — MODELS 2024</b> <ul style="list-style-type: none"> <li>• <i>Fault localization in DSLTrans model transformations by combining symbolic execution and spectrum-based analysis</i> (SoSyM journal-first)</li> </ul>  | Sep. 2024 |
| <b>Invited talk — onto:Nexus Forum 2024</b> <ul style="list-style-type: none"> <li>• <i>Accelerating Digital Twin Construction with Ontologies</i></li> <li>• Forum on Ontological Modeling and Analysis</li> </ul>                                   | Jan 2024  |
| <b>Invited talk — Polytechnique Montréal</b> <ul style="list-style-type: none"> <li>• <i>Bridging the Gap in Verification and Validation of Complex Systems</i></li> </ul>                                                                            | Jan 2023  |
| <b>Guest lecture — LOG6953DE, Polytechnique Montréal</b> <ul style="list-style-type: none"> <li>• Host: Prof. Mohammad Hamdaqa</li> <li>• <i>Model/Graph Transformations: Specification and Verification</i></li> </ul>                               | Nov. 2022 |
| <b>Invited talk — Software Engineering at Montréal (SEMTL) September seminar</b> <ul style="list-style-type: none"> <li>• <i>Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice</i></li> </ul> | Sep. 2022 |
| <b>Invited talk — Concordia University</b> <ul style="list-style-type: none"> <li>• Host: Prof. Essam Mansour</li> <li>• <i>Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice</i></li> </ul>  | Jan 2022  |
| <b>Invited talk — Université du Québec à Montréal</b> <ul style="list-style-type: none"> <li>• <i>Capturing and Utilising Domain-Specific Knowledge</i></li> </ul>                                                                                    | Jan 2022  |

## Community Building

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| <b>Workshop Organizer — onto:Nexus Workshop 2026 (International Workshop on Ontological Modeling and Analysis)</b> <ul style="list-style-type: none"> <li>• Co-located with MODELS 2026, Malaga, Spain.</li> <li>• Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa, Edward Kamburjan</li> </ul>                      | Jan 2026  |
| <b>Workshop Organizer — onto:Nexus Workshop 2025 (International Workshop on Ontological Modeling and Analysis)</b> <ul style="list-style-type: none"> <li>• Co-located with MODELS 2025, Grand Rapids, MI. Hybrid event with IEEE proceedings.</li> <li>• Co-organizers: Maged Elaasar (Chair), Wahab Hamou-Lhadj, Mohammad Hamdaqa</li> </ul> | Oct. 2025 |

**Lead Organizer — Software Engineering at Montréal (SEMTL)**

Montréal, Canada

- Regular seminars for the software engineering researchers in Montréal (~40 attendees per meeting, ~30% are professors).
- Coordinating with meeting hosts on content, venue, date, and maintaining website.

Aug 2022 – present

**Organizing Committee Member**

2022–2026

- SEMLA 2026 — General Chair
- FSE 2026 — Student Volunteer Co-Chair
- MODELS 2025 — Student Volunteer Co-Chair
- ICSE 2025 — Posters Co-Chair
- SEMLA 2024, 2025 — Posters Co-Chair
- EDTconf 2024 — Publicity Co-Chair
- ANNSIM 2024 — Proceedings Co-Chair
- ANNSIM 2022, 2023 — Cyber-Physical Systems Track Co-Chair
- MODELS 2022 — Posters Co-Chair

**Session Chair**

2021–2024

- SEMLA LLM Ops Day — 2024
- Model-Driven Engineering of Digital Twins Workshop (ModDiT) — 2023
- Artificial Intelligence and Model-driven Engineering Workshop (MDEIntelligence) — 2023
- Consortium for Software Engineering Research (CSER) Spring Meeting — 2023
- Model-Driven Engineering and Software Development (MODELSWARD) — 2021

**Program Committee Member**

2019–2026

- ASE Research Track — 2026
- ASE NIER Track — 2025
- MODELS Artifact Evaluation Track — 2026
- MODELS — 2025
- ANNSIM — 2021–2024, 2026
- DIGITA Workshop (PerCom) — 2026
- International Workshop on Architecting and Engineering Digital Twins (AEDT) — 2025, 2026
- EMMSAD — 2025
- EDTconf — 2024, 2025
- MDE Intelligence — 2022, 2023, 2024
- International Workshop on Models and Evolution — 2022
- ACM Student Research Competition — 2022
- Spring/Summer Simulation Conference — 2019, 2020

**Journal Reviewer**

2021–2025

- Journal of Systems & Software (JSS) — 2025
- Information and Software Technology (IST) — 2025
- Journal of Software and Systems Modeling (SoSyM) — 2020, 2021, 2023, 2025 (Top 1% Reviewer 2020, 2021; Best Reviewer Award 2025)
- ACM Transactions on Software Engineering and Methodology (TOSEM) — 2024
- SIMULATION — 2023, 2024 (Best Reviewer Award June 2023)
- Science of Computer Programming (SCP) — 2023
- Journal of Computer Languages (JCL) — 2022
- Journal of Object Technology (JOT) — 2022, 2026
- Empirical Software Engineering (EMSE) — 2022
- IEEE Transactions on Automation Science and Engineering (T-ASE) — 2021

**Guest Editor for Journal Special Issue**

Jan 2023

- SIMULATION — *Modeling and Simulation for Software-Intensive Systems: from IoT to Digital Twins*

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| <b>Jury Member</b>                                                                                                                                                                                                                                                                                                           | 2023–2026    |
| <ul style="list-style-type: none"> <li>• Polytechnique Montréal — PhD Thesis Jury President (2026)</li> <li>• Polytechnique Montréal — M.Sc. Thesis Jury President (2024)</li> <li>• McGill University — PhD Thesis External Reviewer (2024)</li> <li>• McGill University — M.Sc. Thesis External Reviewer (2023)</li> </ul> |              |
| <b>Grant Reviewer</b>                                                                                                                                                                                                                                                                                                        | 2024–2026    |
| <ul style="list-style-type: none"> <li>• NSERC Idea to Innovation grant — 2024</li> <li>• NSERC Discovery grant — 2025, 2026</li> <li>• Dutch Research Council (NWO) — 2026</li> </ul>                                                                                                                                       |              |
| <b>Panelist</b>                                                                                                                                                                                                                                                                                                              | Jan 2023     |
| <ul style="list-style-type: none"> <li>• ANNSIM PhD Colloquium Panel — 2023</li> </ul>                                                                                                                                                                                                                                       |              |
| <b>Departmental Service</b>                                                                                                                                                                                                                                                                                                  | 2025–Present |
| <ul style="list-style-type: none"> <li>• Committee Member, Committee Rayonnement (Communications committee), Department of Computer and Software Engineering, Polytechnique Montréal</li> </ul>                                                                                                                              |              |

## Scholarships and Awards

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| <b>SoSyM Best Reviewer Award</b>                                                                                                         | Jan 2025   |
| <ul style="list-style-type: none"> <li>• Journal of Software &amp; Systems Modeling</li> </ul>                                           |            |
| <b>Simulation Journal Best Reviewer Award</b>                                                                                            | June 2023  |
| <b>SoSyM Top 1% Reviewer</b>                                                                                                             | 2020, 2021 |
| <ul style="list-style-type: none"> <li>• Journal of Software &amp; Systems Modeling</li> </ul>                                           |            |
| <b>Best Student Paper Award</b>                                                                                                          | Jan 2019   |
| <ul style="list-style-type: none"> <li>• SIMULTECH — for the paper <i>HintCO – Hint-based configuration of co-simulations</i></li> </ul> |            |
| <b>NSERC Postgraduate Scholarship – Doctoral (PGS D)</b>                                                                                 | 2015–2016  |
| <ul style="list-style-type: none"> <li>• Natural Sciences and Engineering Research Council of Canada</li> </ul>                          |            |
| <b>Lorne Trottier Science Accelerator Fellowship</b>                                                                                     | 2015, 2016 |
| <ul style="list-style-type: none"> <li>• McGill University</li> </ul>                                                                    |            |
| <b>Harold H. Helm Fellowship</b>                                                                                                         | 2013, 2014 |
| <ul style="list-style-type: none"> <li>• McGill University</li> </ul>                                                                    |            |
| <b>Grad Excellence Award in Computer Science</b>                                                                                         | 2012, 2014 |
| <ul style="list-style-type: none"> <li>• McGill University</li> </ul>                                                                    |            |

## Selected Publications

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| <b>Model-based digital twin engineering: insights, challenges, and future directions</b>                                                                                                                                                                  | May 2026 |
| Philipp Zech, Souvik Barat, Benjamin Nast, <b>Bentley Oakes</b> , Judith Michael, Steffen Zschaler, Balbir Barn, Ruth Breu<br><a href="https://doi.org/10.1007/s10270-026-01368-8">10.1007/s10270-026-01368-8</a> (Software and Systems Modeling (SoSyM)) |          |
| <b>Automated Extraction and Analysis of Developer’s Rationale in Open Source Software</b>                                                                                                                                                                 | Jan 2025 |
| Mouna Dhaouadi, <b>Bentley Oakes</b> , Michalis Famelis<br><a href="https://doi.org/10.1145/3729383">10.1145/3729383</a> (Proceedings of the ACM on Software Engineering (FSE))                                                                           |          |
| <b>DTInsight: A Tool for Explicit, Interactive, and Continuous Digital Twin Reporting</b>                                                                                                                                                                 | Jan 2025 |
| Kérian Fiter, Louis Malassigné-Onfroy, <b>Bentley Oakes</b><br><a href="https://doi.org/10.1109/MODELS-C68889.2025.00030">10.1109/MODELS-C68889.2025.00030</a> (MODELS Companion (MODELS-C))                                                              |          |

- Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State of the Practice** Apr 2024  
**Bentley Oakes**, Michalis Famelis, Houari Sahraoui  
[10.1145/3638243](#) (ACM Transactions on Software Engineering and Methodology (TOSEM))
- Toward Intelligent Generation of Tailored Graphical Concrete Syntax** Jan 2024  
 Meriem Ben Chaaben, Oussama Ben Sghaier, Mouna Dhaouadi, Nafisa Elrasheed, Ikram Darif, Imen Jaoua, **Bentley Oakes**, Eugene Syriani, Mohammad Hamdaqa  
[10.1145/3640310.3674085](#) (MODELS 2024)
- Towards Ontological Service-Driven Engineering of Digital Twins** Jan 2024  
**Bentley Oakes**, Claudio Gomes, Eduard Kamburjan, Giuseppe Abbiati, Elif Ecem Bas, Sebastian Engelsgaard  
[10.1145/3652620.3688261](#) (MODELS Companion (EDTconf) — Best Short Paper Award)
- Toward a Systematic Reporting Framework for Digital Twins: A Cooperative Robotics Case Study** Jan 2024  
 Santiago Gil, **Bentley Oakes**, Cláudio Gomes, Mirgita Frasheri, Peter G. Larsen  
[10.1177/00375497241261406](#) (SIMULATION)
- openCAESAR: Balancing Agility and Rigor in Model-Based Systems Engineering** Jan 2023  
 Maged Elaasar, Nicolas Rouquette, David Wagner, **Bentley Oakes**, Abdelwahab Hamou-Lhadj, Mohammad Hamdaqa  
[10.1109/MODELS-C59198.2023.00051](#) (MODELS Companion (MODELS-C))
- Machine Learning-Based Fault Injection for Hazard Analysis and Risk Assessment** Jan 2021  
**Bentley Oakes**, Mehrdad Moradi, Simon Van Mierlo, Hans Vangheluwe, Joachim Denil  
[10.1007/978-3-030-83903-1\\_12](#) (SAFECOMP 2021 (Computer Safety, Reliability, and Security))

## Publications

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- See [bentleyoakes.com/publications](https://bentleyoakes.com/publications) for a full list of dozens of peer-reviewed publications, including journal articles, book chapters, conference papers, workshop papers, and technical reports/theses.